THE PUSHFORWARD AND THE GLOBAL SECTIONS OF THE PULLBACK OF A MODULE UNDER THE STRUCTURE MORPHISM IN THE VANISHING SITUATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Quasi-coherence of the pushforward of a quasi-coherent module. Morphisms of Spaces, label spaces-morphisms-lemma-pushforward (invoked at proof line 3019). It shows that $p_*(M \otimes_A \mathcal{O}_X)$ is quasi-coherent on $\Spec(A)$, so that the equality of global sections determines it as $\widetilde{M}$.
- Global sections of direct sums of the structure sheaf. Same chapter spaces-cohomology.tex, label lemma-colimits (invoked at proof line 3022). It gives $\Gamma(X, \bigoplus_{i \in I}\mathcal{O}_X) = \bigoplus_{i \in I}A$, which is the free case of the claim.
- Vanishing of H^1 in the situation, which makes global sections right exact. Same chapter spaces-cohomology.tex, Situation \ref{situation-vanishing} and the vanishing statement of the surrounding section (used at proof line 3025). It lets the global sections of a presentation of M compute the global sections of $M \otimes_A \mathcal{O}_X$.

Source reference. Original lemma 073E: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 667–667. Navigation pointer for \ref{lemma-colimits} used inside the counted span (line 3022); the argument that consumes it is retained in the span.

Source reference. Original reference 07UY: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2991–2991. Navigation pointer for \ref{situation-vanishing} used inside the counted span (line 3010); the argument that consumes it is retained in the span.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03M9: sources/stacks-chow/tags.txt, source lines 4741–4741. Navigation pointer for \ref{spaces-morphisms-lemma-pushforward} used inside the counted span (line 3020); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 07UY for an A-module M we have $p_*(M \otimes_A \mathcal{O}_X) = \widetilde{M}$ and $\Gamma(X, M \otimes_A \mathcal{O}_X) = M$.*

Proof. The equality $p_*(M \otimes_A \mathcal{O}_X) = \widetilde{M}$ follows from the equality $\Gamma(X, M \otimes_A \mathcal{O}_X) = M$ as $p_*(M \otimes_A \mathcal{O}_X)$ is a quasi-coherent module on $\mathrm{Spec}(A)$ by Morphisms of Spaces, Lemma 03M9. Observe that $\Gamma(X, \bigoplus_{i \in I} \mathcal{O}_X) = \bigoplus_{i \in I} A$ by Lemma 073E. Hence the lemma holds for free modules. Choose a short exact sequence $F_1 \to F_0 \to M$ where F_0, F_1 are free A-modules. Since $H^1(X, -)$ is zero the global sections functor is right exact. Moreover the pullback p^* is right exact as well. Hence we see that

$$\Gamma(X, F_1 \otimes_A \mathcal{O}_X) \to \Gamma(X, F_0 \otimes_A \mathcal{O}_X) \to \Gamma(X, M \otimes_A \mathcal{O}_X) \to 0$$

is exact. The result follows. □